Fiona Reza

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Coastal Environmental Researcher | Aquatic Cultivation | Water-Quality Monitoring | Environmental Data Analysis

Education

M.S. in Oceanography, Norfolk, VA
Old Dominion University Fall 2024 – Fall 2026

M.S. in Oceanography, Dhaka, Bangladesh
University of Dhaka 2020-2022

B.S. in Oceanography, Dhaka, Bangladesh
University of Dhaka 2016-2020

Research Experience

Graduate Research, Old Dominion University 2024 - 2026
Growth Conditions and Mixotrophic Behavior of Harmful algae (*Alexandrium monilatum)*

- Investigating effects of temperature, salinity, light, and nitrogen sources on the growth and potential bloom.
- Designing controlled lab culture experiments using different salinity and temperature treatments and quantifying with microscopy, FlowCam, and fluorometry-based monitoring.
- Developing mixotrophic feeding assays to test cryptophyte prey effects.

Research Assistant, Old Dominion University Summers 2025 & 2026
Seasonal Shifts of Phytoplankton Assemblages in the Lafayette River

- Investigating seasonal phytoplankton shifts and bloom dynamics using weekly to high-frequency observations linked with environmental and nutrient variability.
- Collecting field samples and water-quality data, including temperature, salinity, dissolved oxygen, tidal depth, nutrients, chlorophyll a, particulate nitrogen, and particulate carbon.
- Quantifying major plankton groups using FlowCam to assess biomass and community responses to salinity, temperature, oxygen, and nutrient availability.

Master's Thesis, Department of Oceanography, University of Dhaka 2021 - 2022
Extraction and Characterization of Sodium Alginate from Brown Seaweeds

- Proposed local brown seaweeds as sustainable sources for sodium alginate production.
- Compared $CaCl_2$ and formaldehyde-based extraction methods using FTIR, viscosity, and molecular-weight analysis.
- Demonstrated higher alginate yield and quality from *Sargassum* sp. and *Padina* sp.

Field Experience

Plankton and Benthic Sampling Demonstration, Elizabeth River, Virginia, Old Dominion University 2024–2026

- Demonstrated plankton net collection and benthic grab sampling during teaching lab and student field sessions.
- Assisted students with basic sample handling, observation, and field-based marine sampling procedures.

Brown Seaweed Field Collection, St. Martin's Island, Bangladesh 2021–2022

- Collected brown seaweed during winter and spring for alginate extraction, characterization, and species identification.

Marine Water-Quality Sampling, Swatch of No Ground, Bay of Bengal, Bangladesh 2020

- Collected field samples for chlorophyll-a, nitrate, phosphate, ammonia, temperature, and salinity analysis from Bangladesh's deepest marine canyon region.

Laboratory & Technical Skills

Plankton, Algal Culture & Microscopy: Microscopy, FlowCam, PlanktoScope, algal culture maintenance, media preparation, incubator setup
Water-Quality & Biogeochemical Analysis: YSI probe deployment, nutrient analysis, ammonium autoanalyzer, fluorometry, spectrophotometry, refractometry
Analytical & Chemical Laboratory Techniques: Chemical handling, sodium alginate extraction, rolling-ball viscometry, FTIR, NMR
Environmental & Optical Instrumentation: Quantum scalar laboratory radiometer
GIS & Mapping: ArcGIS, ERDAS Imagine
Programming: Python, MATLAB
Data & Scientific Software: OriginLab, ChemDraw

Teaching Experience

Teaching Assistant, Introduction to Oceanography Lab (Fall & Springs) 2024–2026

- Led three weekly lab sections for undergraduate students, demonstrating oceanographic concepts, experiments, and data-analysis activities.
- Graded lab notebooks, homework, and exams while helping students with scientific writing, hypothesis development, and experiment interpretation.
- Assisted instructional cruises involving water, sediment, salinity, temperature, and nutrient sample collection.

Freelance Math and Science Tutor 2021–2023

- Tutored 20+ students in grades 8–10 in math and science, supporting homework, exam preparation, and concept review.

Publications

1. Islam, M. S., Ahmed, M. K., Chowdhury, K. M. A., Islam, M. S., Shampa, M. T. A., Reza, F., & Albin, E. "A Preliminary Approach to Investigate the Abundance, Spatial Distribution, and Ecological Risk of Microplastics in Surface Water and Sediment of the Naf River, Bangladesh." *Journal of Coastal Conservation*. (Manuscript Under Review)

Poster Presentations

1. Reza, F., Momtaz, N., et al. "Production of Sodium Alginate from Naturally Occurring Seaweeds in Bangladesh and Its Prospects in Industrial Biotechnology." *International Conference on Biotechnology in Sustainable Development*, 2023.
2. Reza, F., Momtaz, N., et al. "Physicochemical Characterization of Sodium Alginate and Evaluation of the Industrial Value of Common Alginophytes from St. Martin's Island, Bangladesh." *International Conference on Environmental Protection for Sustainable Development (ICEPSD)*, 2022.
3. Reza, F., Ahmed, M. K., Chowdhury, K. M. A., & Munshi, J. L. "Characterization of Sodium Alginate and Evaluation of the Industrial Value of Common Alginophytes from St. Martin's Island, Bangladesh." *Career Festival and Research Fair, Faculty of Earth and Environmental Sciences, University of Dhaka*, 2022.
4. Abdullah, M., Ahmed, M. K., Khandakar, M. M., & Reza, F. "Plankton Diversity and Abundance Concerning Biophysical Conditions Around Kutubdia Island, East Coast of Bangladesh." *4th International Conference on Bangladesh Environment (ICBEN-4)*, 2020.

Scholarships and Awards

NST Fellowship 2022
Awarded by the Ministry of Science and Technology, Bangladesh.

Best Oral Presenter Award 2022
International Conference on Environmental Protection for Sustainable Development (ICEPSD).

Pre-Graduation Scholarship 2016–2020
Awarded by the University of Dhaka; included full free studentship and monthly stipend.

Volunteering Experience

Measure the Muck Floodwater Sampling, Norfolk, Virginia. 2024

Blue Crab Bowl, Room Runner
High school science competition, Virginia Institute of Marine Science, Gloucester Point, Virginia. 2024

Conference Volunteer
International Conference on Earth and Environmental Sciences & Technology for Sustainable Development (ICEEST). 2020